

Module: **Problem 2 - Dealing with Hands**

This module provides utilities for manipulating a “hand” of letters in word-game style applications.

A hand is represented as a dictionary that maps each letter (a single-character string) to the count of how many times that letter is available.

The primary operation is to consume letters from a hand when a word is played, returning a new hand without mutating the original.

Functions

`updateHand(hand, word)`

Description:

Creates a new hand dictionary that reflects the consumption of the letters required to form `word`.

For each character in `word`, the corresponding count in the hand is decremented; if a count reaches zero, that entry is removed from the hand.

The original `hand` dictionary is left unchanged (no side effects).

Parameters: - `hand` (*dict[str, int]*): A mapping of letters to their available counts. The function assumes that the hand contains at least as many of each letter as appears in `word`. - `word` (*str*): The word whose letters should be removed from the hand.

Returns:

`dict[str, int]` — a new hand dictionary with the used letters removed (or reduced in count).

Examples:

```
# Crash examples observed during testing
result = updateHand(0, 'hello')    # → AttributeError: 'int' object has no attribute 'copy'
result = updateHand(1, 'test')     # → AttributeError: 'int' object has no attribute 'copy'
result = updateHand(-1, '')        # → AttributeError: 'int' object has no attribute 'copy'
```

Edge Cases / Notes: - **Invalid `hand` type:** Passing a non-dictionary (e.g., an `int`) for `hand` triggers an `AttributeError` because the implementation calls `hand.copy()`. Ensure `hand` is a dictionary before invoking the function. - **Missing letters:** If `word` contains a letter not present in `hand` (or more occurrences than available), a `KeyError` will be raised when the code attempts to decrement a non-existent entry. - **Empty word:** An empty string results in the hand being returned unchanged, provided `hand` is a valid dictionary. - **No side effects:** The original `hand` object is never modified; a shallow copy is created and returned.